Remote Workforce & Employee Well-being Analytics

PHASE 1- DATA PREPARATION

1.FIRST I READ THE EXCEL FILE ,MAKE SURE THAT EVERYTHING IS CORRECT.

2.NEXT I LOAD THAT CSV FILE IN POWERBI WHERE I CLICK THAT TRANSFORM DATA TO OPEN POWER QUERY

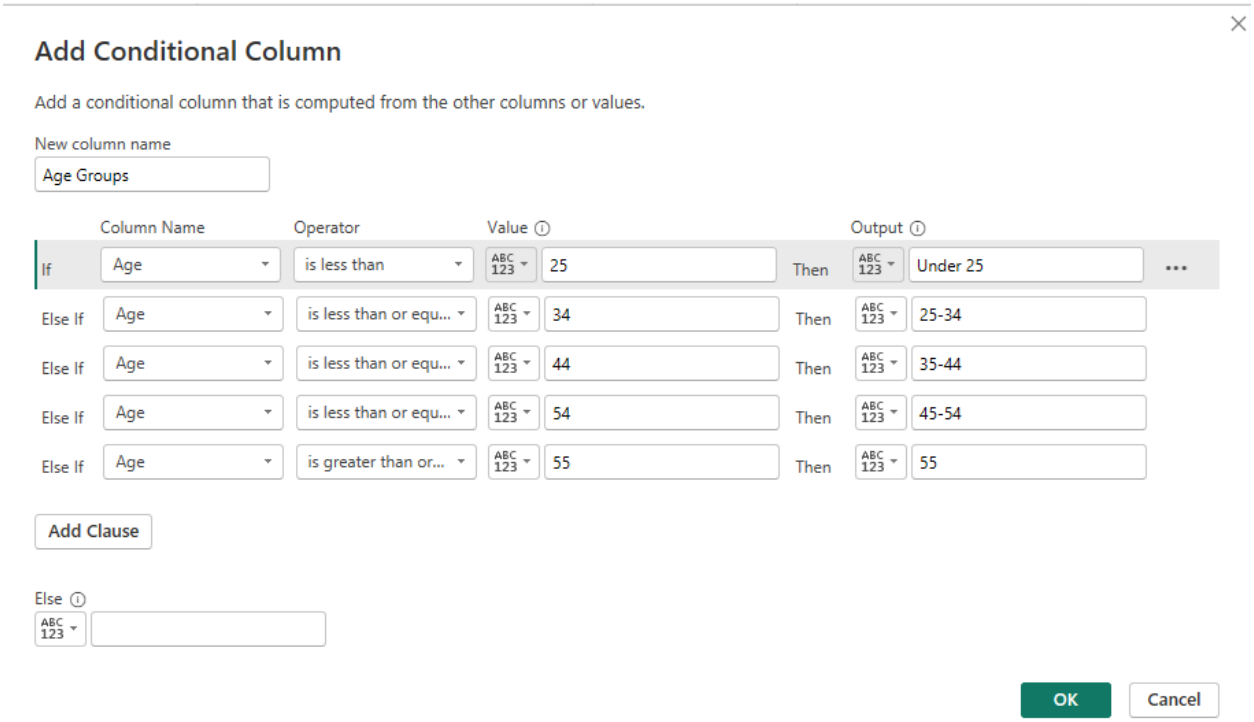
3.AFTER THAT I CHECKED THE DATA TYPE OF ALL COLUMNS AND THERE IS NO MISMATCH

4. I CHECKED FOR MISSING OR DUPLICATES VALUES IN EMPLOYEE ID COLUMN THERE IS NONE

5.I CHECKED ALL EMPTY ROWS FROM COLUMNS STATISTICS .ALSO THERE IS NO EMPTY ROW

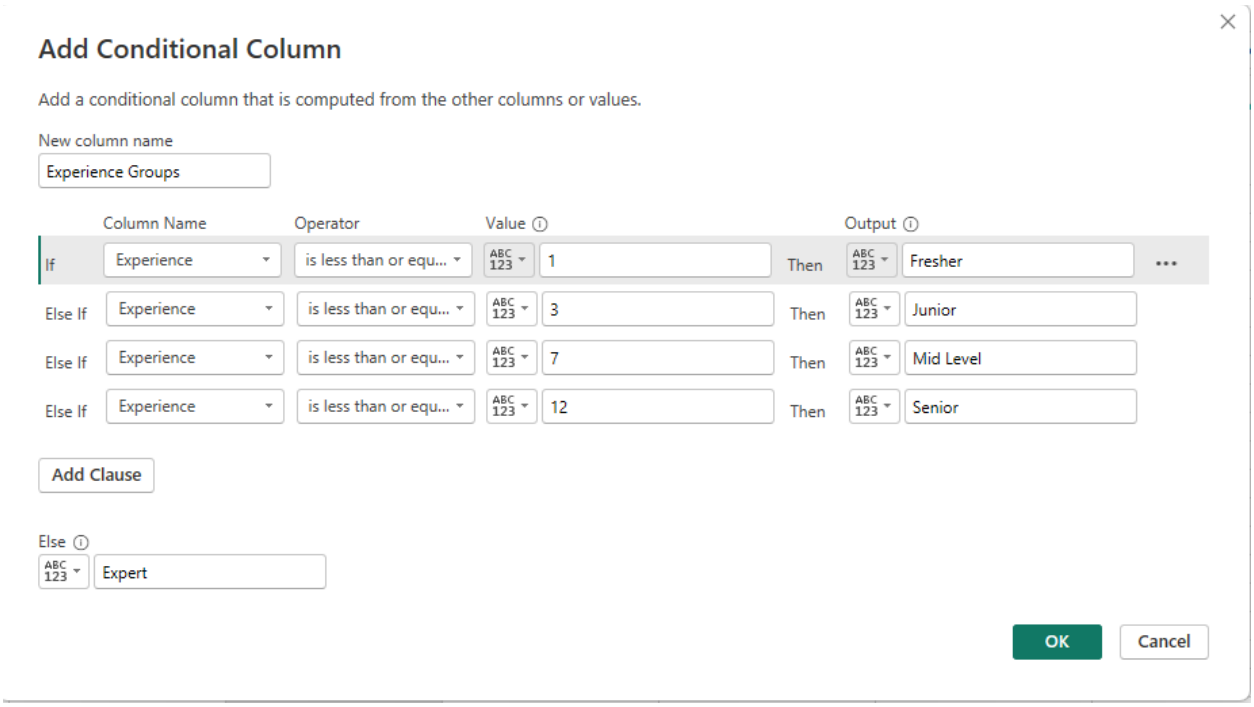
6.AFTER THAT I CHANGE ALL THE COLUMN NAME FROM LONG FORM TO SHORT FORM WITHOUT CHANGING THE MEANING AND I REMOVED THE “-” SIGN CAUSE IT LOOK UNPROFESSIONAL.

7.AFTER THIS I CREATED AGE GROUPS BY ADDING CONDITIONAL COLUMN AND PUT VALUES WHERE NEEDED TO CREATE THAT GROUP. LIKE THIS-



LIKE THIS ALL AGE GROUP I CREATED

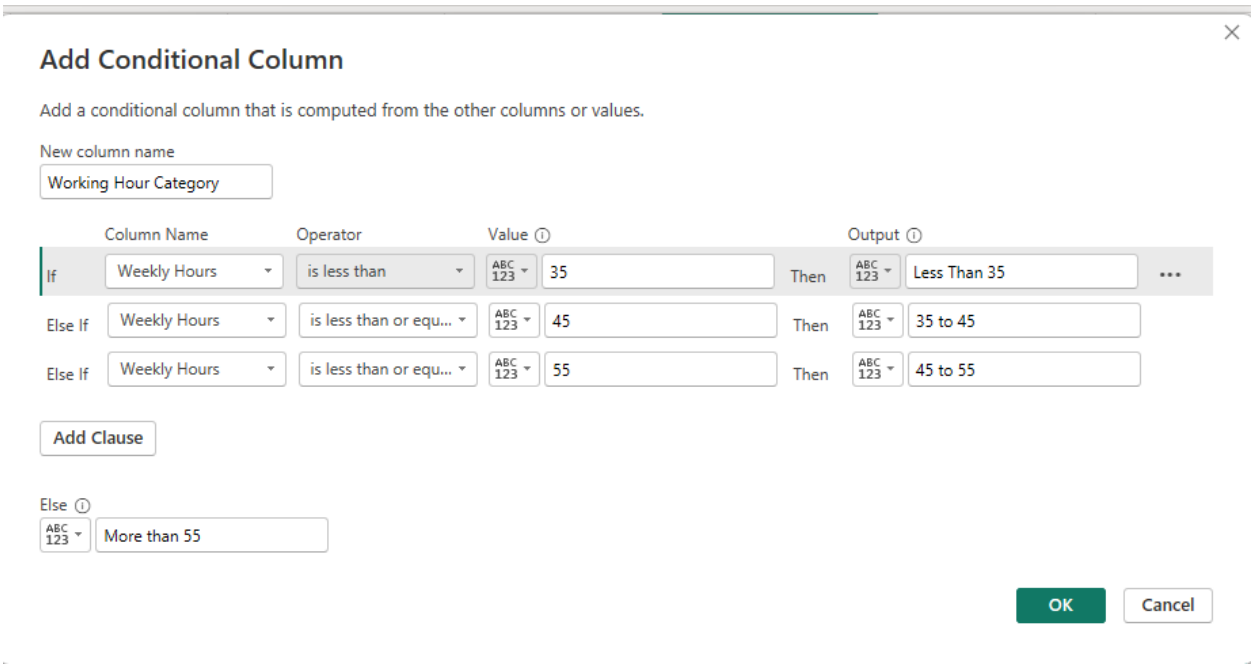
8.AFTER THIS EXPERIENCE GROUPS NEED TO BE CREATED



I CREATED THIS GROUP AS FRESHER, JUNIOR,MID-LEVEL, SENIOR AND EXPERT BASED LIKELY ON THEIR EXPERIENCE

I TAKE LESS THAN 1 YEAR EXPERIENCE AS FRESHER ,1 TO 3 AS JUNIOR ,3 TO 7 AS MID-LEVEL ,7 TO 12 AS SENIOR AND ABOVE 13 AS EXPERT .

9.AFTER THIS WEEKLY HOURS ALSO NEED TO BE CATEGORIZED SAME AS EXPERIENCE LIKE THIS -



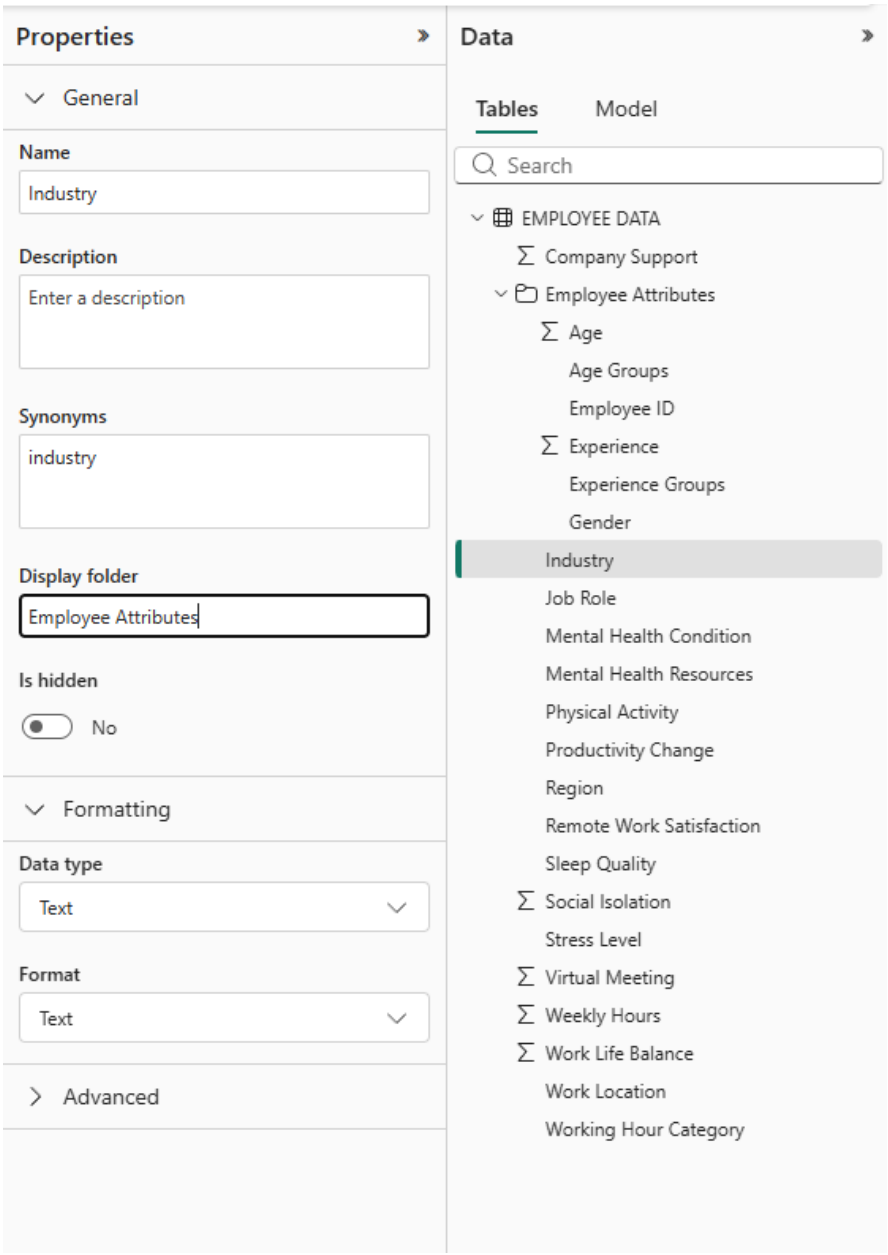
MAINLY AS LESS THAN 35, 35 TO 45, 45 TO 55 AND MORE THAN 55. WEEKLY HOURS CATEGORY ALSO GET CREATED.

SO ALL NECESSARY WORK IS DONE IN POWER QUERY AND I CLOSE IT AND APPLY

PHASE 2 – DATA MODELLING

1.SO THE FIRST THING I DO IS I CHANGE THE ORIGINAL NAME OF THE FILE TO EMPLOYEE DATA AS IT WILL BE EASY TO UNDERSTAND AND ORGANISE THE DATA

2.AFTER THIS I ORGANIZE THE DATA IN MODEL VIEW IN SEPARATE FOLDER CONTAINING SAME ATTRIBUTES LIKE EMPLOYEE INFORMATION IN EMPLOYEE ATTRIBUTES FOLDER -



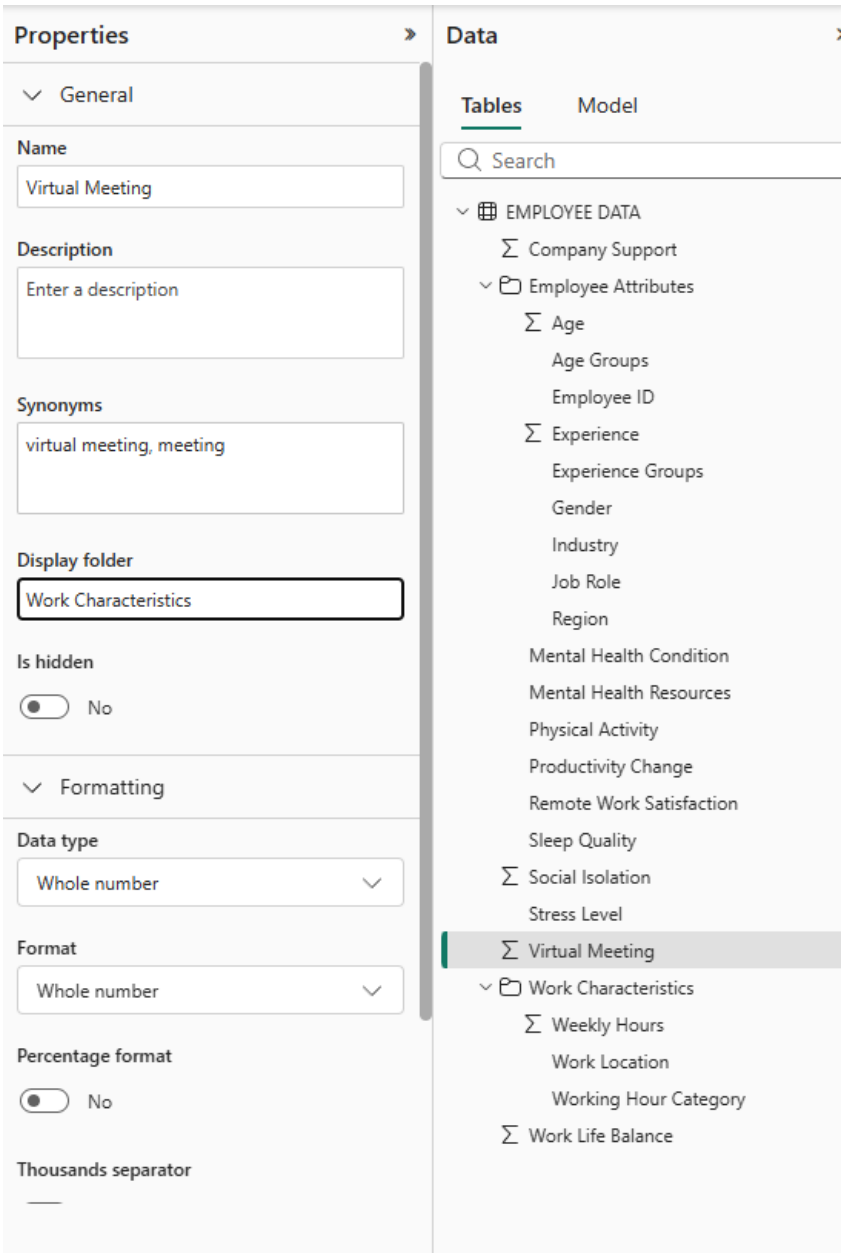
- Age Groups
- Employee ID
- Experience
- Experience Groups
- Gender
- Industry
- Job Role

- Region

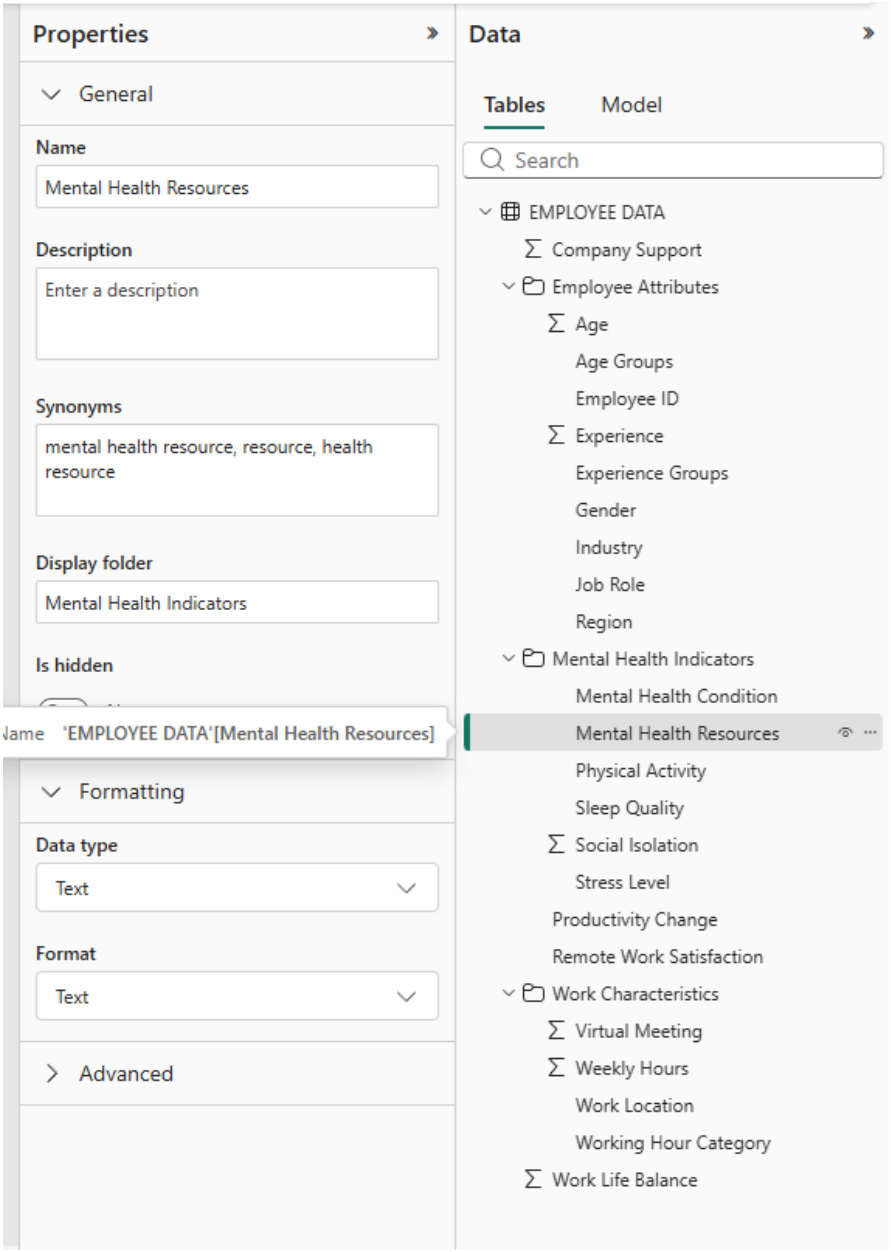
THIS ALL COMES UNDER THE EMPLOYEE ATTRIBUTES DISPLAY FOLDER

3.AFTER THIS I HAVE DONE SAME FOR WORK CHARACTERISTICS PUT ALL WORK RELATED CHARACTERISTICS IN SINGLE FOLDER IT INCLUDES

- Work Location
- Weekly Hours
- Working Hour Category
- Virtual Meeting



4.SAME GOES FOR MENTAL HEALTH INDICATORS ALSO LIKE-



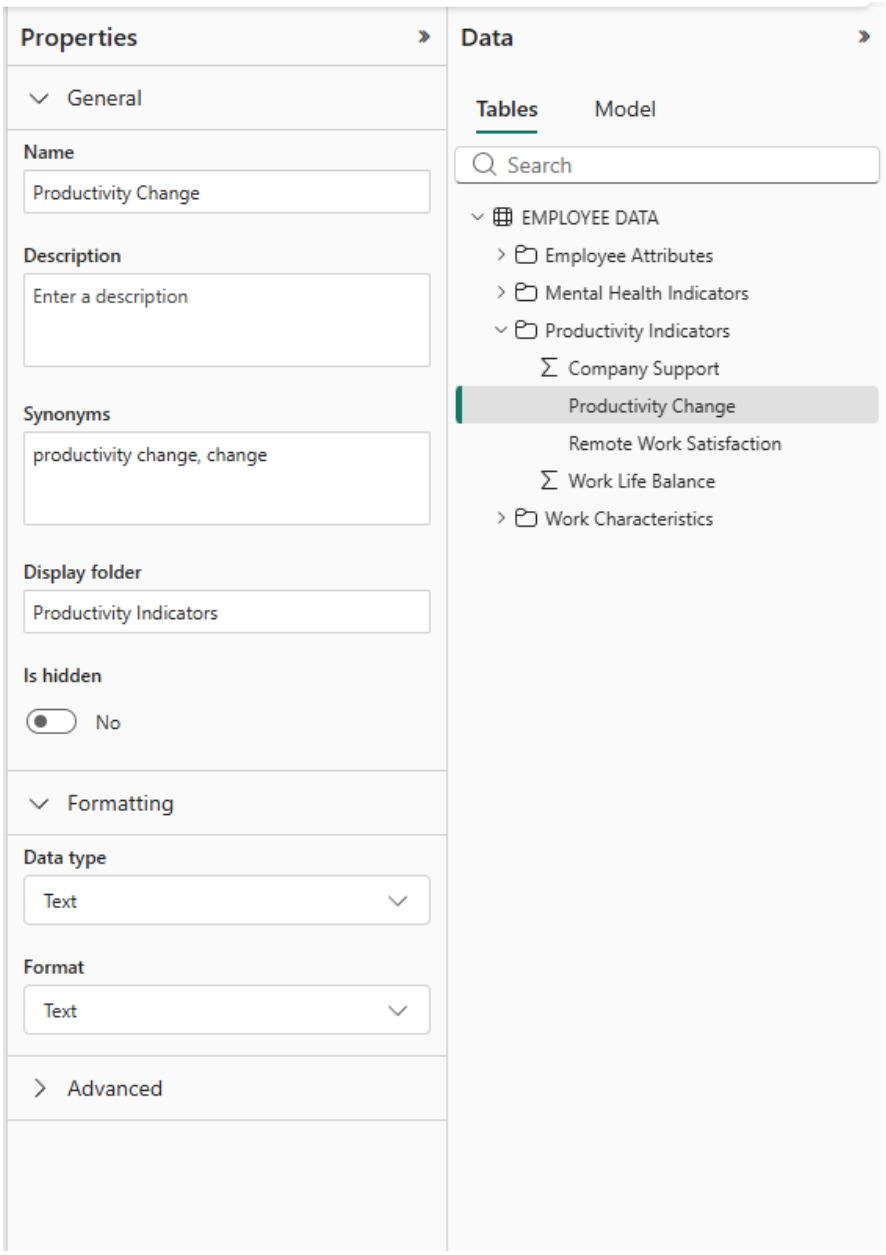
IT INCLUDE ALL MENTAL RELATED ACTIVITY

- Mental Health Condition
- Mental Health Resources
- Stress Level
- Social Isolation
- Sleep Quality
- Physical Activity

5.SAME THING HERE AT PRODUCTIVITY INDICATORS DISPLAY FOLDER

IT INCLUDE THIS THINGS

- Productivity Change
- Work Life Balance
- Remote Work Satisfaction
- Company Support



OK NOW ALL THE DATA IS ORGANISED

PHASE 3- DAX CALCULATIONS

BASIC MEASURES-

1.SO WE CREATED OUR FIRST DAX WHICH IS TOTAL EMPLOYEES AND TO SEPARATE THEORIGINAL COLUMNS AND THE MEASURES WE ARE CREATING WE ARE GONNA PUT THE NEW MEASURES IN NEW FOLDER THAT IS MEASURES FOLDERS.

```
Total Employees = DISTINCTCOUNT('EMPLOYEE DATA'[Employee ID])
```

2.AFTER THAT I CREATED AVERAGE AGE MEASURE AND PUT IT IN MEASURES FOLDER

```
Average Age = AVERAGE('EMPLOYEE DATA'[Age])
```

3.SAME GOES FOR AVERAGE EXPERIENCE MEASURES

```
Average Experience = AVERAGE('EMPLOYEE DATA'[Experience])
```

4.AGAIN SAME FORMULA FOR AVERAGE WORKING HOURS

```
Average Working Hours = AVERAGE('EMPLOYEE DATA'[Weekly Hours])
```

5.SAME FOR VIRTUAL MEETING AVERAGE

```
Average Virtual Meetings = AVERAGE('EMPLOYEE DATA'[Virtual Meetings])
```

6.AVERAGE WORK-LIFE BALANCE

```
Average Work-Life Balance =
AVERAGE('EMPLOYEE DATA'[Work Life Balance])
```

7.Average Remote Work Satisfaction

```
Average Remote Work Satisfaction =
```

AVERAGE('EMPLOYEE DATA'[Remote Work Satisfaction])

8. AVERAGE SLEEP QUALITY SAME AS ABOVE

Average Sleep Quality =

AVERAGE('EMPLOYEE DATA'[Sleep Quality])

9. AVERAGE SOCIAL ISOLATION

Average Social Isolation =

AVERAGE('EMPLOYEE DATA'[Social Isolation])

10.AVERAGE COMPANY SUPPORT RATING

Average Company Support =

AVERAGE('EMPLOYEE DATA'[Company Support])

HERE THE BASIC MEASURES FOLDERS WORK IS DONE NOW WE WILL MOVE TO EMPLOYEE DISTRIBUTION MEASURES

EMPLOYEE DISTRIBUTION MEASURES -

1.EMPLOYEES BY GENDER-

For male-

Male Employees =

CALCULATE(

[Total Employees],

'EMPLOYEE DATA'[Gender] = "Male"

)

For female-

```
Female Employees =
CALCULATE(
   [Total Employees],
   'EMPLOYEE DATA'[Gender] = "Female"
)
```

2.EMPLOYEE BY WORK LOCATION-

```
REMOTE=
Remote Employees =
CALCULATE(
   [Total Employees],
   'EMPLOYEE DATA'[Work Location] = "Remote"
)
HYBRID=
Hybrid Employees =
CALCULATE(
   [Total Employees],
   'EMPLOYEE DATA'[Work Location] = "Hybrid"
)
ONSITE=
On-site Employees =
CALCULATE(
   [Total Employees],
   'EMPLOYEE DATA'[Work Location] = "On-site"
)
```

3.Employees By Region-

```
Asia=
Asia Region=
CALCULATE(
   [Total Employees],
   'EMPLOYEE DATA'[Region] = "Asia"
)
AFRICA=
Africa Region=
CALCULATE(
   [Total Employees],
   'EMPLOYEE DATA'[Region] = "Africa"
)
EUROPE=
Europe Region=
CALCULATE(
   [Total Employees],
   'EMPLOYEE DATA'[Region] = "Europe"
)
SOUTH AMERICA=
South America Region=
CALCULATE(
   [Total Employees],
   'EMPLOYEE DATA'[Region] = "South America"
)
NORTH AMERICA=
```

North America Region=

```
CALCULATE(
   [Total Employees],
   'EMPLOYEE DATA'[Region] = "North America"
)
```

OCEANIA=

Oceania Region=

```
CALCULATE(
   [Total Employees],
   'EMPLOYEE DATA'[Region] = " Oceania "
)
```

4.Employees By Job Role-

HR-

HR Employees =

```
CALCULATE(
   [Total Employees],
   'EMPLOYEE DATA'[Job Role] = "HR"
)
```

Designer-

Designer Employees =

```
CALCULATE(
   [Total Employees],
   'EMPLOYEE DATA'[Job Role] = " Designer "
)
```

Data Scientist-

```
Data Scientist =
CALCULATE(
   [Total Employees],
   'EMPLOYEE DATA'[Job Role] = " Data Scientist "
)
```

Sales-

```
Sales Employees =
CALCULATE(
   [Total Employees],
   'EMPLOYEE DATA'[Job Role] = " Sales "
)
```

Project Manager -

```
Project Manager Employees =
CALCULATE(
   [Total Employees],
   'EMPLOYEE DATA'[Job Role] = " Project Manager "
)
```

Software Engineer -

```
Software Engineer Employees =
CALCULATE(
   [Total Employees],
   'EMPLOYEE DATA'[Job Role] = " Software Engineer "
```

)

Marketing-

Marketing =

CALCULATE(

[Total Employees],

'EMPLOYEE DATA'[Job Role] = " Marketing "

)

SO HERE JOB ROLE IS ALSO OVER AND WE ALSO PUT IT ALL IN EMPLOYEE DISTRIBUTION MEASURES

5.EMPLOYEES BY INDUSTRY-

1.IT INDUSTRY-

IT INDUSTRY=

CALCULATE(

[Total Employees],

'EMPLOYEE DATA'[Industry] = " IT "

)

2.Manufacturing INDUSTRY-

Manufacturing INDUSTRY=

CALCULATE(

[Total Employees],

'EMPLOYEE DATA'[Industry] = " Manufacturing "

)

3.Education INDUSTRY-

```
Education INDUSTRY=
CALCULATE(
   [Total Employees],
   'EMPLOYEE DATA'[Industry] = " Education "
)
```

4.Consulting INDUSTRY-

```
Consulting INDUSTRY=
CALCULATE(
   [Total Employees],
   'EMPLOYEE DATA'[Industry] = " Consulting "
)
```

5.Retail INDUSTRY-

```
Retail INDUSTRY=
CALCULATE(
   [Total Employees],
   'EMPLOYEE DATA'[Industry] = " Retail "
)
```

6.Healthcare INDUSTRY-

```
Healthcare INDUSTRY=
CALCULATE(
   [Total Employees],
   'EMPLOYEE DATA'[Industry] = " Healthcare "
```

)

7.Finance INDUSTRY-

Finance INDUSTRY=

CALCULATE(

[Total Employees],

'EMPLOYEE DATA'[Industry] = " Finance "

)

3.Productivity Measures

1.Employees With Increased Productivity

Increased Productivity Employees =

CALCULATE(

[Total Employees],

'EMPLOYEE DATA'[Productivity Change] = "Increase"

)

2.Employees With Decreased Productivity

Decreased Productivity Employees =

CALCULATE(

[Total Employees],

'EMPLOYEE DATA'[Productivity Change] = "IDecrease"

)

3.Employees With No Productivity Change

No Productivity Change Employees =

CALCULATE(

[Total Employees],

'EMPLOYEE DATA'[Productivity Change] = "No Change"

)

4.Productivity Improvement Percentage-

Productivity Improvement Percentage =

DIVIDE(

[Increased Productivity Employees],

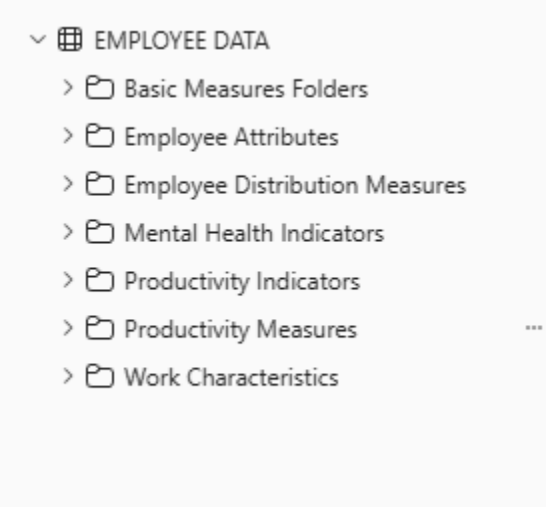
[Total Employees],

0

) * 100

HERE WE USE (*100) TO GET THE NUMBER IN PERCENTAGE

HERE IS THE QUICK UPDATE WHAT WE HAVE DONE -



WE KEEP ALL OUR MEASURE ORGANIZED AND CLEAN FOR MORE READIBILITY

SO NOW WE ARE CREATING MENTAL HEALTH MEASURES

4.MENTAL HEALTH MEASURES

1.HIGH STRESS EMPLOYEES-

```
High Stress Employees =
CALCULATE(
   [Total Employees],
   'EMPLOYEE DATA'[Stress Level] = "High"
)
```

2.LOW STRESS EMPLOYEES-

```
High Stress Employees =
CALCULATE(
   [Total Employees],
   'EMPLOYEE DATA'[Stress Level] = "Low"
)
```

3.MEDIUM STRESS EMPLOYEES-

```
High Stress Employees =
CALCULATE(
   [Total Employees],
   'EMPLOYEE DATA'[Stress Level] = “ Medium"
)
```

4.EMPLOYEES WITH MENTAL HEALTH CONDITION-

```
Employees with Mental Health Conditions =
CALCULATE(
   [Total Employees],
   'EMPLOYEE DATA'[Mental Health Condition] = "Yes"
)
```

5.EMPLOYEES WITH MENTAL HEALTH RESOURCES

```
Employees with Mental Health Resources =
CALCULATE(
   [Total Employees],
   'EMPLOYEE DATA'[Mental Health Resources] = "Yes"
)
```

6.ACCESS PERCENTAGE

```
Access Percentage =
DIVIDE(
   [Employees with Mental Health Resources],
   [Total Employees],
   0
)
```

HERE THE MENTAL HEALTH MEASURES IS ALSO COMPLETED NEXT SATISFACTION MEASURES

5. SATISFACTION MEASURES-

1.AVERAGE SATISFACTION-

```
Average Satisfaction =
AVERAGE('EMPLOYEE DATA'[Remote Work Satisfaction])
```

2.AVERAGE WORKLIFE BALANCE-

```
Average Satisfaction =
```

AVERAGE('EMPLOYEE DATA'[Remote Work Satisfaction])

3.AVERAGE COMPANY SUPPORT-

Avg Company Support =

AVERAGE('EMPLOYEE DATA'[Company Support])

4.AVERAGE SLEEP QUALITY-

Avg Sleep Quality =

AVERAGE('EMPLOYEE DATA'[Sleep Quality])

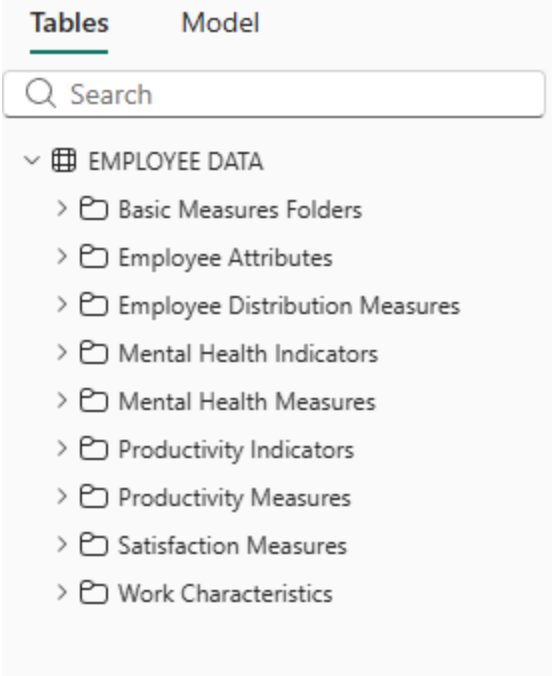
5.AVERAGE PHYSICAL ACTIVITY-

Average Physical Activity =

AVERAGE('EMPLOYEE DATA'[Physical Activity])

SO OUR SATISFACTION MEASURES IS ALSO COMPLETE.

CURRENTLY IT LOOK LIKE THIS



BY THIS ALL OUR MEASURES CALCULATIONS IS DONE .

Phase 4 – Dashboard 1: Executive Overview

FIRST WHEN I TRY TO BUILD THE KPI I CAME ACROSS THE PROBLEM THAT THE AVERAGE I CREATED EARLIER IS ON CATEGORICAL BASIS MEANS THE CONTENT IN CELL IS IN TEXT FORMAT LIKE INCREASE,DECREASE,NO CHANGE NOT IN SCORE BASIS.SO WHEN I CAME ACROSS THIS PROBLEM , I CREATED AND OTHER COLUMNS ON NUMERICAL BASIS WHICH INCLUDE ALL PRODUCTIVITY SCORE,SATISFACTION SCORE,SLEEP QUALITY SCORE AND STRESS SCORE AND TO DIFFERENCIATE BETWEEN OTHER MEASURES I PUT THAT ALL INTO SEPARATE DISPLAY FOLDER THAT IS 'NUMERICAL MEASURES'.SO THIS IS THE PROBLEM .

DASHBOARDS

1.EXECUTIVE OVERVIEW

KPI'S-

- Total Employees
- Average Work-Life Balance
- Average Stress Level (mapped numerically)

• Average Satisfaction

• Average Sleep Quality

• Average Productivity Score (mapped numerically)

AND FOR VISUALS I M CREATING-

1.EXECUTIVE OVERVIEW DASHBOARD

• Employees by Region (Map or Filled Map if region supports it)

• Employees by Work Location (Donut Chart)

• Employees by Gender (Pie/Donut Chart)

• Employees by Industry (Bar Chart)

• Employees by Job Role (Bar Chart)

2.PRODUCTIVITY ANALYTICS

Business Questions

• Which work location has the highest productivity?

• Does experience affect productivity?

• Does the number of meetings impact productivity?

Suggested Visuals

• Productivity Change by Work Location

• Productivity Change by Industry

• Productivity by Job Role

• Productivity by Experience Group

• Productivity by Hours Worked

• Productivity by Virtual Meetings

3 – Employee Well-being

Business Questions

- Which employees report the highest stress?
- Does work-life balance reduce stress?
- Does sleep quality impact stress?
- Which regions experience poor work-life balance?

Suggested Visuals

- Stress Level Distribution
- Stress by Region
- Stress by Job Role
- Work-Life Balance by Work Location
- Sleep Quality by Stress Level
- Physical Activity by Stress Level

Dashboard 4 – Mental Health Analytics

Business Questions

- How many employees have access to mental health resources?
- Does access improve satisfaction?
- Does access reduce stress?

Suggested Visuals

- Access to Resources (Donut Chart)

• Satisfaction by Resource Access

• Stress by Resource Access

• Mental Health Condition by Region

• Mental Health Condition by Job Role

Dashboard 5 – Remote Work Analysis

Business Questions

• Which work location provides the best work-life balance?

• Which work location has the happiest employees?

• Which work location receives the highest company support?

Suggested Visuals

• Satisfaction by Work Location

• Company Support by Work Location

• Work-Life Balance by Work Location

• Productivity by Work Location

• Stress by Work Location

Dashboard 6 – HR Insights

Create an executive page answering:

Which regions have:

• Highest stress?

• Lowest satisfaction?

• Highest productivity?

• Lowest company support?

Which job roles have:

• Highest stress?

• Lowest work-life balance?

• Highest social isolation?

• Lowest sleep quality?

DRILL THROUGH PAGES

I ADD NEW PAGE AND NAME THAT PAGE REGION AFTER THAT I PUT REGION ON DRILL THROUGH OPTION AND ADD OTHER KPI CARDS SO THAT DRILL THROUGH OPRION WILL WORK WELL

SAME GOES FOR OTHER TWO DRILL THROUGH PAGE THAT IS JOB ROLE DETAILS AND INDUSTRY DETAILS

SLICER INSERTION AND INTERACTIVITY-

ADDED REGION,INDUSTRY ,WORK LOCATION,GENDER,JOB ROLE,STRESS LEVEL

I ADD ALL THE SLICERS DO SOME SYNCING FOR ALL PAGES

DYNAMIC TITLE-

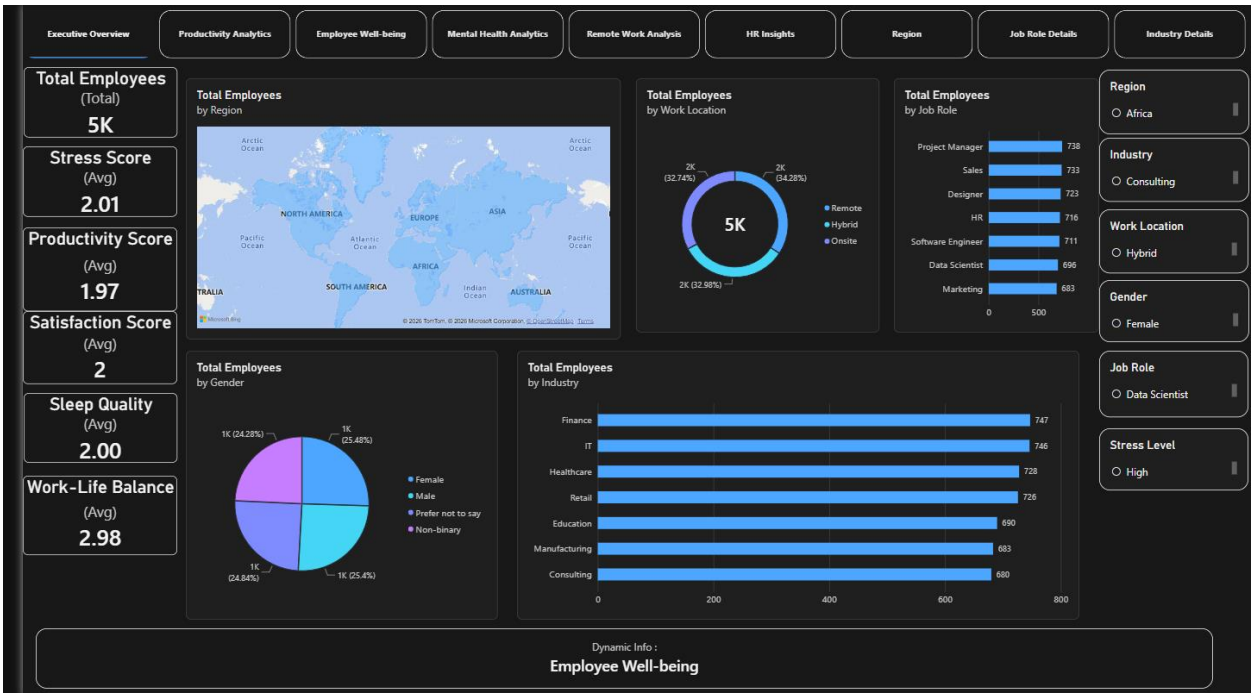
I NEXT ADD DYNAMIC TITLE WHERE I CREATED AN MEASURE WHICH INCLUDE ALL 6 SLICERS WHICH ARE SYNCED TO EACH OTHER.NOW WHEN WE TAP ON ANY SLICER ALL THE PAGES GET FORMATTED TO THAT PARTICULAR COMMAND

NAVIGATION-

AFTER DYNAMIC TITLE I ADD NAVIGATION BUTTONS FOR BETTER FUCTIONALITY AND READIBILITY

AFTER THIS ALL FUNCTIONALITY ITS TIME FOR FORMATING.

DONE WITH FORMATTING ADD SOME BLACK THEME.



THIS IS THE LAYOUT I CHOOSE

ALL SLICER , FUNCTIONALITIES ,CHARTS ,VISUAL,CROSS REPORTING IS WORKING.

AFTER THIS I ADD TOOLTIP FOR EMPLOYEE WELLBEING .

AND AT LAST I CREATED HEIRERCHY FOR REGION,INDUSTRY,JOB ROLE AND ALSO ADD BOOKMARKS ADDING ALL THE TASK THAT IS MENTION IN OUR PDF.

Business Questions & Answers

1. **What percentage of employees work remotely, hybrid, and on-site?**
   Answer: Remote: **[34.28%]**, Hybrid: **[32.98%]**, On-site: **[32.74%]**.

2. **Which work location has the highest employee satisfaction?**
   Answer: Remote, Hybrid, and On-site employees have the **same average employee satisfaction score**, so no work location has a higher satisfaction level than the others.

3. **Which job roles experience the highest stress levels?**
   Answer: Project Manager experiences the highest average stress level, with a score of 14.76% And 738 Count.

4. **Does higher work-life balance correlate with increased productivity?**
   Answer: No clear positive correlation is observed. The analysis does not show that employees with higher work-life balance consistently have higher productivity.

5. **Which regions have the greatest social isolation?**
   Answer: Oceania has the highest average social isolation score of **[X]**.

6. **Does access to mental health resources improve employee satisfaction?**
   Answer: No. Employees **without access** to mental health resources report slightly higher average satisfaction (**2.01**) compared with employees **with access** (**1.99**). Therefore, access to mental health resources does **not appear to improve employee satisfaction** in this dataset.

7. **How do working hours affect stress and work-life balance?**
   Answer: Employees working **more than 55 hours** show the **highest stress** (average stress score **2.05**), while employees working **35–45 hours** show the **lowest work-life balance** (average WLB **2.9**).

8. **Does sleep quality appear to influence productivity?**
   Answer: Employees with **Sleep Quality 2 and 3** show the highest average productivity score of **1.97**, compared with **1.96** for Sleep Quality 1. The difference is minimal, so sleep quality does not appear to strongly influence productivity in this dataset.

9. **Which industries report the highest stress?**
   Answer: Education reports the highest average stress score of **2.05**.

10. **Which experience group has the highest satisfaction?**
    Answer: Fresher has the highest average satisfaction score of **2.05**.

11. **Which work location receives the strongest company support?**
    Answer:Remote work location receives the highest average company support score of **3.04**.

12. **Which regions should HR prioritize for wellness initiatives?**
    Answer: HR should prioritize **Europe**, as it records the **highest average stress (2.04)**, relatively **low employee satisfaction (2.02)**, and the **lowest company support (2.96)** among the regions. **Asia** may also require attention due to its lowest satisfaction score (1.97).